

Modeling Mental Health Treatment-Seeking Behavior to Inform Workplace Support Strategies

Workplace mental health is a significant population health challenge that affects both individual well-being and organizational productivity. In the United States, approximately one in five adults experiences a mental health condition, with nearly half affected over their lifetime [2]. These conditions contribute to substantial economic burden, with projected productivity losses reaching \$6 trillion by 2030 [1], highlighting the need for scalable, prevention-focused strategies.

Despite this impact, many employees do not seek care due to stigma, limited awareness, and insufficient workplace support. These barriers reflect broader organizational and social determinants of health, including workplace culture, access to resources, and perceived professional risk. Evidence suggests that supportive organizational practices—such as leadership engagement and stigma reduction—can improve both employee well-being and organizational outcomes [3].

This study applies a data-driven population health approach guided by the CDC Health Impact in Five Years (HI-5) framework. Using the OSMI Mental Health in Tech Survey dataset, classification and clustering techniques are employed to identify individuals at risk of not seeking care and to segment the workforce based on barriers to treatment. The results aim to inform targeted, evidence-based interventions and support a scalable pipeline with continuous monitoring to improve workplace mental health outcomes.

Literature Review (Adil)

Employee mental health has emerged as a central concern for both organizational researchers and practitioners, given its profound impact on well-being, productivity, and long-term workforce outcomes. According to the National Institute of Mental Health, 1 in 5 adults in the U.S. are affected by mental health conditions and roughly 50% of people will experience a mental health disorder during their lifetime [2]. The impact of mental health is not isolated to only employees but employers as well. Estimates show a \$6 trillion accrued loss to employers by 2030 from productivity dips related to employees impacted by mental health conditions [1].

Given the above, It is mutually beneficial for both the working class and their employers to drive towards an achievement of better mental health across the workforce. Although studies show a consensus of positive impacts for all parties when striving towards improving mental health outcomes for the working class population (and the general population), the difficult part is identifying how to get this population the proper treatments and, perhaps even more importantly, understanding mental health seeking behavior to ensure the treatments are utilized. To begin, we must explore the social determinants of mental health in the working class population.

Mental health help-seeking behavior is influenced by a variety of social determinants, including stigma, socioeconomic status, and cultural norms. Research shows that stigma remains one of the most significant barriers preventing individuals from accessing professional care [4]. Additionally, social class may influence attitudes toward treatment, particularly among working-class populations where cultural expectations of self-reliance can discourage seeking psychological support [5].

Furthermore, generational differences in attitudes toward mental health may influence help-seeking patterns. Older generations may be more likely to view psychological distress as a personal issue that should be handled privately, contributing to lower utilization of mental health services. In contrast, younger populations tend to demonstrate higher levels of mental health literacy and greater openness toward counseling or therapy [6].

Once the intricacies of the working population is understood and segmented, workplace-based intervention methods can be developed and offered to employees with the goal of ensuring that mental health seeking behaviors are maximized for each cohort. Examples of intervention methods include mental health education modules, stress management training, mindfulness and anxiety-reducing techniques, and employee assistance programs (EAPs). A study performed on employees from a Malaysian manufacturing company implemented many of these methods and found that employees who participated in these programs experience significant reductions in depression, stress, and anxiety levels after the intervention period [7].

While most interventions are performed on the employee level, employers can take an alternative route of attempting to indirectly impact an employees wellbeing by targeting the leaders or workplace benefits for intervention instead, with the goal of curating a better workplace environment. Interventions here would include mental health management training, promotion of work-life balance, and increasing paid time off or adopting mental health days. Evidence has shown an emphasis on cultivating a psychologically healthy workplace through supportive leadership and an offering of employee-sponsored mental health resources that lead to favorable outcomes related to employee health and well-being [3].

Literature Review (Tania)

Workplace mental health is a critical concern for both employees and organizations, as it directly influences well-being, productivity, and long-term workforce outcomes. Evidence indicates that supportive workplace policies, access to mental health resources, and psychologically supportive environments are associated with improved employee outcomes and greater engagement with care [8, 9, 10, 11]. In contrast, limited organizational support and workplace stigma remain key barriers to addressing mental health needs, contributing to reduced productivity, increased absenteeism, and higher employee turnover [8, 9].

Understanding help-seeking behavior requires examining both individual and systemic determinants. Social and organizational factors such as stigma, workplace culture, and structural support strongly influence whether employees access mental health resources [11, 12]. Stigma is consistently identified as a major barrier, particularly in environments that emphasize performance and self-reliance [11, 12]. Broader workplace guidance also highlights the importance of organizational context, access to services, and awareness in shaping mental health behaviors [12, 13].

Given this complexity, segmenting the workforce based on risk factors and barriers is essential for effective intervention design. The literature supports combining individual-level strategies—such as mental health education, stress management, mindfulness programs, and Employee Assistance Programs—with organizational-level approaches, including leadership training, supportive workplace culture, and improved access to resources [8, 9, 15]. Evidence

from systematic reviews shows that multilevel interventions integrating both approaches are more effective in improving mental health outcomes and workplace functioning [9, 10, 11].

Overall, workplace mental health interventions are most effective when they integrate individual support with organizational change while addressing systemic determinants of health. These approaches align with public health and workplace guidelines that emphasize prevention, access, and sustainability [12, 13, 14]. This study builds on this evidence by applying data-driven methods to identify at-risk employees and segment populations, enabling targeted and scalable interventions within a population health framework [12, 14].

Dataset Description

The dataset used in this project is the **OSMI Mental Health in Tech Survey (2014)**, collected by Open Sourcing Mental Illness and publicly available on Kaggle [2]. This survey is among the few publicly accessible datasets that captures **employee mental health attitudes, employer support behaviors, and treatment-seeking patterns** in a single instrument, providing a rich resource for population-level analysis [3, 7].

The dataset contains **1,259 respondents** and **27 features** across three primary domains:

1. **Demographics** – including age and gender.
2. **Work environment** – encompassing company size, remote work policies, employer-provided benefits, anonymity protections, and leave policies.
3. **Mental health perceptions** – including family history, work interference, fear of professional consequences, and comfort discussing mental health with coworkers and supervisors.

The **target variable**, *treatment*, captures whether respondents sought mental health treatment and is naturally balanced at **51% Yes and 49% No**, making it well-suited for binary classification tasks without the need for resampling [7].

Prior to analysis, several data cleaning steps were applied. Non-informative columns such as Timestamp, Comments, State, and Country were removed. Age values were filtered to a realistic working range of 18–75 years, eliminating erroneous entries. The Gender column initially contained 49 unique free-text responses, which were standardized into three categories: Male, Female, and Non-binary. Missing values in the *work_interfere* column were imputed as "N/A" to distinguish between employees with no mental health condition and those whose condition did not interfere with work. Additionally, missing values in *self_employed* were filled with the column mode.

The final cleaned dataset consists of **1,251 rows and 24 columns with no missing values**, providing a robust and reliable foundation for classification and clustering analyses aimed at understanding mental health treatment-seeking behavior in the workplace [7, 11].

Data Analysis & Results

Our approach to this analysis was to use both classification and clustering methods on this dataset with the goal of applying two separate interventions based on each model's output. We

will discuss intervention methods in the next section but below is are the results of both methods we applied:

Classification Results:

Before exploring and comparing different classification models to apply onto this data, a target variable had to be created. The goal when developing the target variable was to identify employees who had a mental health condition but were not actively seeking help. After experimenting with a few variable conditions, we landed on the target variable being flagged as employees who responded “Sometimes” or “Often” to the work_interfere variable and responses of “No” to the seek_help variable. The reason we chose these two variables is because the seek_help question attempts to address whether the employer offers any benefits to assist, which is useful in our target variable since our goal is to introduce interventions so any “No” responses allow us to target employees who feel they do not have adequate benefits.

We chose to pair this variable response with the work_interfere variable because the question was framed in a way where strong responses (“Sometimes” and “Often”) meant that the employees did believe they had a mental health condition and that work did interfere in some degree (the actual question was: If you have a mental health condition, do you feel that interferes with your work?). Below is the distribution of the two questions and their responses and the bold portion shows the sum/percentage of those flagged true in relation to the target variable:

work_interfere	Total	Won't Seek Help	Rate
Never	477	227	47.6%
Rarely	173	79	45.7%
Sometimes	465	247	53.1%
Often	144	93	64.6%

Now that we have developed the target variable, we decided to test three separate classification models on the data: Random Forest, Decision Trees, and Gradient Boost. The results of the three models are summarized below:

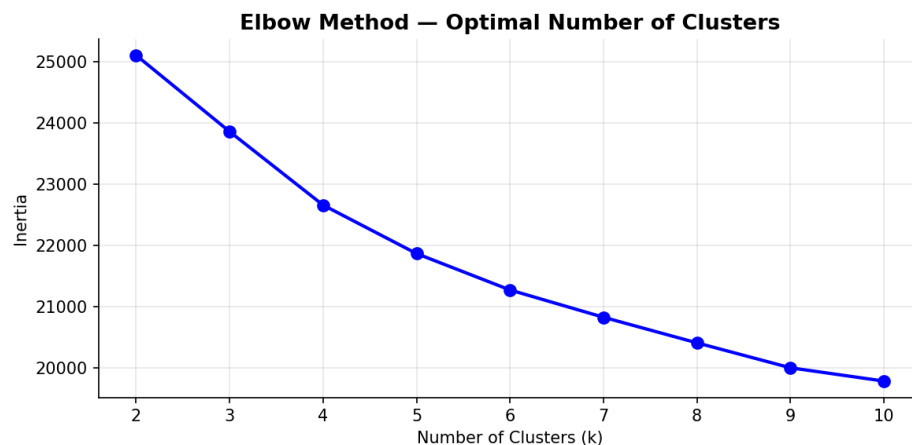
Model	CV AUC	Test AUC	Avg Precision	Opt. Recall	Opt. Precision	Employees Flagged
Logistic Regression	0.896	0.899	0.724	0.926	0.573	110
Decision Tree	0.946	0.967	0.905	0.912	0.747	83
Random Forest	0.957	0.967	0.91	0.941	0.771	83
Gradient Boosting	0.958	0.965	0.892	0.912	0.775	80

The highest scores are highlighted in yellow for each measurement. Our logistic regression model was the weakest of the bunch as it ranked last in all measurements and also flagged the most employees, which will impact the efforts needed in our interventions. The other three models performed closely to each other but there are some nuanced differences between them.

Cross Validation and Test AUC scores were all above 94% for the Decision Tree, Random Forest, and Gradient Boosting models. However, our focus is on the Recall metrics to ensure we are prioritizing limiting the amount of false negatives because missing someone who needs outreach and support is our biggest concern. That being said, the highest Optimized Recall score comes from the Random Forest model which also had a solid Optimized Precision score and reasonable amount of employees flagged compared to the other models.

Clustering Results:

In addition to classification, we applied K-Means clustering to segment the workforce into groups based on similarities in demographics, work environment, and mental health perceptions. All non-numeric features were label-encoded, and feature values were standardized prior to clustering to ensure equal weighting in distance calculations. The elbow method was used to determine the optimal number of clusters, with the curve flattening at $k = 4$, indicating four distinct groups as a balance between granularity and interpretability.



The four clusters were profiled to understand the characteristics and barriers to treatment-seeking for each group. Key cluster statistics are summarized below:

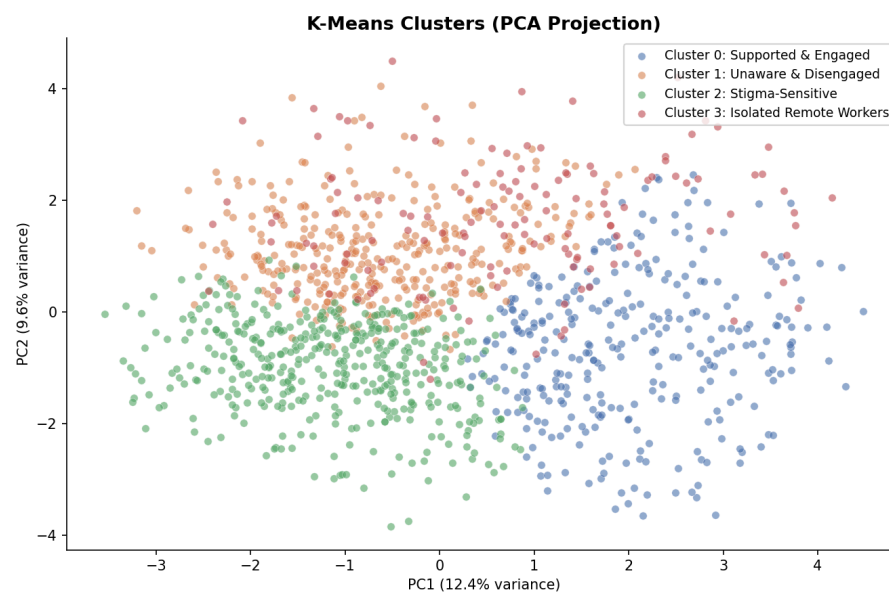
- **Cluster 0: Supported & Engaged (n = 309)**
High levels of treatment-seeking (68.6%), strong employer-provided benefits (83.2%), and high anonymity protection (72.2%). Most employees were on-site (82.5%) and reported moderate work interference (56.3%). This cluster reflects employees who are aware of, and supported by, workplace resources.
- **Cluster 1: Unaware & Disengaged (n = 369)**
Low treatment rate (36.6%) with minimal employer support (benefits 16.8%, seek_help 4.9%) and low anonymity protection (15.4%). This cluster largely comprises on-site employees (68%) who may be disengaged or unaware of available mental health resources.
- **Cluster 2: Stigma-Sensitive (n = 434)**
Moderate treatment-seeking (48.6%) but high fear of professional consequences (42.9%) and very low anonymity protection (8.3%). Employees in this cluster experience

moderate work interference (48.4%) and limited employer support, suggesting stigma and perceived workplace risk as barriers to seeking care.

- **Cluster 3: Isolated Remote Workers (n = 139)**

Moderate treatment-seeking (53.2%) with low employer support (benefits 13.7%, seek_help 14.4%) but predominantly remote (71.2%). High levels of work interference (58.3%) indicate that isolation and lack of direct workplace engagement may contribute to underutilization of mental health resources.

A PCA visualization of the four clusters confirms separation and distinct profiles in reduced two-dimensional space. A comparative analysis of key metrics across clusters highlights significant differences in treatment rates, work interference, and perceived employer support, supporting a data-driven segmentation approach for targeted interventions.



The cluster-based segmentation provides actionable insights for tailoring interventions to employee subgroups with distinct barriers, including stigma, lack of awareness, and remote work isolation. This complements the classification analysis by identifying groups of employees who may benefit from community- or group-level interventions, rather than individually-targeted approaches.

Proposed Interventions

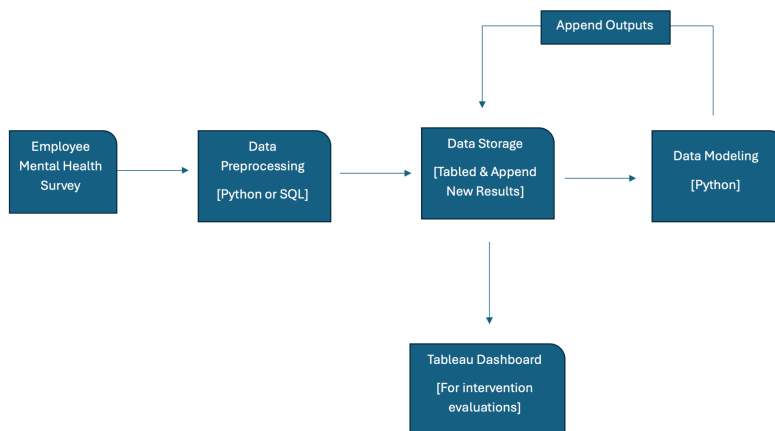
Classification Intervention:

Based on the model outputs, we recommend company-wide interventions targeting the entire employee population through a top-down approach. The primary audience includes HR decision-makers and the managers of any employees who completed the survey.

We propose two interventions. The first is an expansion of the company's mental health benefits. This would involve partnering with HR to evaluate current offerings and identify improvements, such as Employee Assistance Programs (EAPs), digital mental health platforms like Lyra Health or Spring Health, and dedicated mental health days. The specific recommendations would depend on the company's existing benefits and budget, and would require further discussion with HR to determine feasibility.

The second intervention requires mental health training for managers whose direct reports were flagged by the model — the 83 employees identified as having a mental health condition but not seeking help. Training would cover recognizing signs of mental health struggles, best practices for addressing them, and knowing when to escalate a concern. It could also focus more broadly on fostering a supportive work environment, given that workplace conditions are a well-documented contributor to mental health challenges. The format and frequency of training would be determined in collaboration with HR.

To support these interventions, employee survey responses would be fed into the production model, with outputs flowing into an internal Tableau dashboard for HR review. The dashboard would serve three purposes: allowing HR to examine flagged individuals' responses and identify patterns to inform better benefit design; aggregating results by manager and department to pinpoint areas with the highest concentration of unaddressed mental health needs; and tracking trends over time to evaluate whether interventions are improving survey outcomes. The frequency of this analysis would depend on how often the survey is administered. Below is an example of the data pipeline model that could be used to support ongoing interventions:



Clustering Intervention:

Using the K-Means clustering results, we propose data-driven, population-level interventions tailored to employee subgroups based on mental health barriers, work environment, and access to benefits. Unlike the classification intervention, which targets individuals flagged as at-risk, the clustering approach enables segmented, population health strategies that address shared barriers within each group.

The primary audience includes Human Resources (HR) leaders, managers, and team leads. These interventions align with CDC population health competencies by leveraging data to assess risk

(Competency 1), addressing workplace and social determinants such as stigma and access (Competency 2), and applying targeted prevention strategies (Competency 5). The secondary target audience includes employees within each identified cluster, as interventions are designed to address population-level barriers specific to each subgroup.

Proposed Cluster-Specific Interventions:

1. **Supported & Engaged (Cluster 0):** Employees in this group have high access to benefits and actively engage with mental health resources. Interventions should reinforce positive behaviors through wellness programs, peer support initiatives, and continued access to services, with HR monitoring participation to sustain engagement.
2. **Unaware & Disengaged (Cluster 1):** This cluster demonstrates low awareness and utilization of mental health resources. Interventions should focus on targeted awareness campaigns, mental health literacy workshops, and clear guidance on accessing available benefits, with ongoing tracking of engagement to improve uptake.
3. **Stigma-Sensitive (Cluster 2):** Employees in this group express concerns about stigma, confidentiality, and workplace consequences, leading to reduced help-seeking. Interventions should emphasize confidential support options such as anonymous counseling and teletherapy, alongside manager training to reduce stigma and promote a psychologically safe environment.
4. **Isolated Remote Workers (Cluster 3):** These employees primarily work remotely and may have limited access to peer support and workplace resources. Interventions should provide virtual mental health services, remote peer support networks, and regular manager check-ins to reduce isolation and encourage engagement.

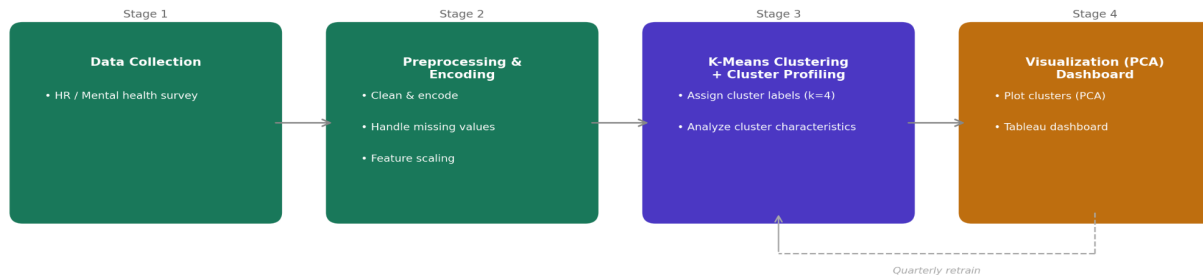
These interventions align with the CDC Health Impact in Five Years (HI-5) framework by addressing multiple levels of population health. At the community level, outreach and awareness initiatives improve mental health literacy and engagement. At the system level, employer policies, manager training, and workplace culture changes address structural barriers such as stigma and access. At the clinical level, expanded services including telehealth, counseling, and Employee Assistance Programs (EAPs) support early intervention and care. Together, this multi-level approach promotes increased treatment-seeking behavior and improved mental health outcomes across employee populations.

Data Pipeline for Clustering Intervention:

A scalable data pipeline was developed to support continuous monitoring and refinement of employee mental health insights. It begins with HR survey data collection, followed by preprocessing steps including data cleaning, encoding categorical variables, handling missing values, and feature scaling. K-Means clustering is then applied to segment employees into distinct groups, followed by cluster profiling to interpret key characteristics and barriers within each segment. Results are visualized using a PCA-based Tableau dashboard, enabling HR and organizational leaders to explore cluster distributions and track trends over time. The pipeline is designed for periodic retraining to ensure outputs remain relevant as new data becomes available, supporting adaptive, data-driven interventions.

This pipeline aligns with CDC Population Health Competencies by using survey data to assess population mental health (Competency 1), identifying social and workplace determinants such as stigma, access, and work environment (Competency 2), incorporating individual- and

population-level risk indicators (Competency 3), applying evidence-based intervention strategies (Competency 4), supporting prevention-focused approaches (Competency 5), leveraging organizational resources such as EAPs and telehealth (Competency 6), and promoting engagement through peer support and manager-led outreach to reduce stigma and improve access to care (Competency 7).



Conclusion

This study demonstrates that mental health improvements in a working population can be advanced through data driven methods including the classification and clustering methods applied onto a population's survey data. In the classification case - we leveraged Random Forest to identify 83 at-risk employees at a high recall to ensure we prioritize a model that limited the number of false negatives. K-Means clustering further revealed four clustered employee profiles: Supported & Engaged, Unaware & Disengaged, Stigma-Sensitive, and Isolated Remote Workers. These two models were leveraged for interventions that included a top-down approach (leadership training and benefit enhancements) and bottom-up approach (employee targeted communications), respectively.

While these interventions were developed based on the analysis from these two models, it's important to note that the final step applying these would require the business leaders to agree to the interventions and also have the resources available to implement things like training, benefit offerings, and targeted communications.

Ultimately, our work underscores that improving mental health in the workplace is both a priority for employees and a business imperative for employers. By integrating predictive analytics with public health principles, companies can start becoming more proactive in their support leading to a reduction of productivity losses and fostering a more supportive mental health workplaces that benefits employees and employers alike.

References:

1. Taubman, D.S., Parikh, S.V. Understanding and Addressing Mental Health Disorders: a Workplace Imperative. *Curr Psychiatry Rep* 25, 455–463 (2023).
2. National Institute of Mental Health. (n.d.). *Mental illness*. U.S. Department of Health and Human Services, National Institutes of Health. Retrieved [Month Day, Year], from

3. Wu, A., Roemer, E. C., Kent, K. B., Ballard, D. W., & Goetzel, R. Z. (2021). *Organizational best practices supporting mental health in the workplace*. *Journal of Occupational and Environmental Medicine*, 63(12), e925–e931.
4. Gulliver A, Griffiths KM, Christensen H. Perceived barriers and facilitators to mental health help-seeking in young people: A systematic review. *BMC Psychiatry*. 2010;10:113.
5. Mahalik JR, Dagirmanjian F. Working-class men's constructions of help-seeking when feeling depressed or sad. *American Journal of Men's Health*. 2019;13(3).
6. Mackenzie CS, Gekoski WL, Knox VJ. Age, gender, and the underutilization of mental health services: The influence of help-seeking attitudes. *Aging & Mental Health*. 2006;10(6):574-582.
7. Mohamed AF, Isahak M, Awg Isa MZ, Nordin R. The effectiveness of workplace health promotion program in reducing work-related depression, anxiety and stress among manufacturing workers in Malaysia. *International Archives of Occupational and Environmental Health*. 2022;95(5):1113-1127
8. Belita, E., Neil-Sztramko, S. E., Seale, C., et al. *Organizational interventions to support and promote the mental health of healthcare workers during pandemics and epidemics: a systematic review*. *BMC Health Services Research*. 2025;25(731).
9. Wagner, S. L., Koehn, C., White, M. I., et al. *Mental health interventions in the workplace and work outcomes: a best-evidence synthesis of systematic reviews*. *International Journal of Occupational and Environmental Medicine*. 2016;7(1):1–14.
10. Joyce, S., Modini, M., Christensen, H., Mykletun, A., Bryant, R., Mitchell, P. B., & Harvey, S. B. *Workplace interventions for common mental disorders: a systematic meta-review*. *Psychological Medicine*. 2016;46(4):683–697.
11. NIOSH/CDC. *Supporting mental health in the workplace*. NIOSH Science Bulletin. U.S. CDC. 2024.
12. World Health Organization. *Mental health at work: fact sheet*. WHO. 2024.
13. NICE (National Institute for Health and Care Excellence). *Mental wellbeing at work: Guidelines*. NICE Guideline No. 212. 2022.
14. Vinson, D. W., Arcan, M., Niland, D-P., & Delahunty, F. *Towards sustainable workplace mental health: a novel approach to early intervention and support*. (Preprint) 2024.
15. *Improving employee well-being and effectiveness: systematic review and meta-analysis of web-based psychological interventions delivered in the workplace*. *Journal of Medical Internet Research*. 2017;19(7):e271.
16. *Effectiveness and feasibility of workplace-based mental health interventions for university academic staff: a systematic review*. *Int J Environ Res Public Health*. 2025;22(12):1787.